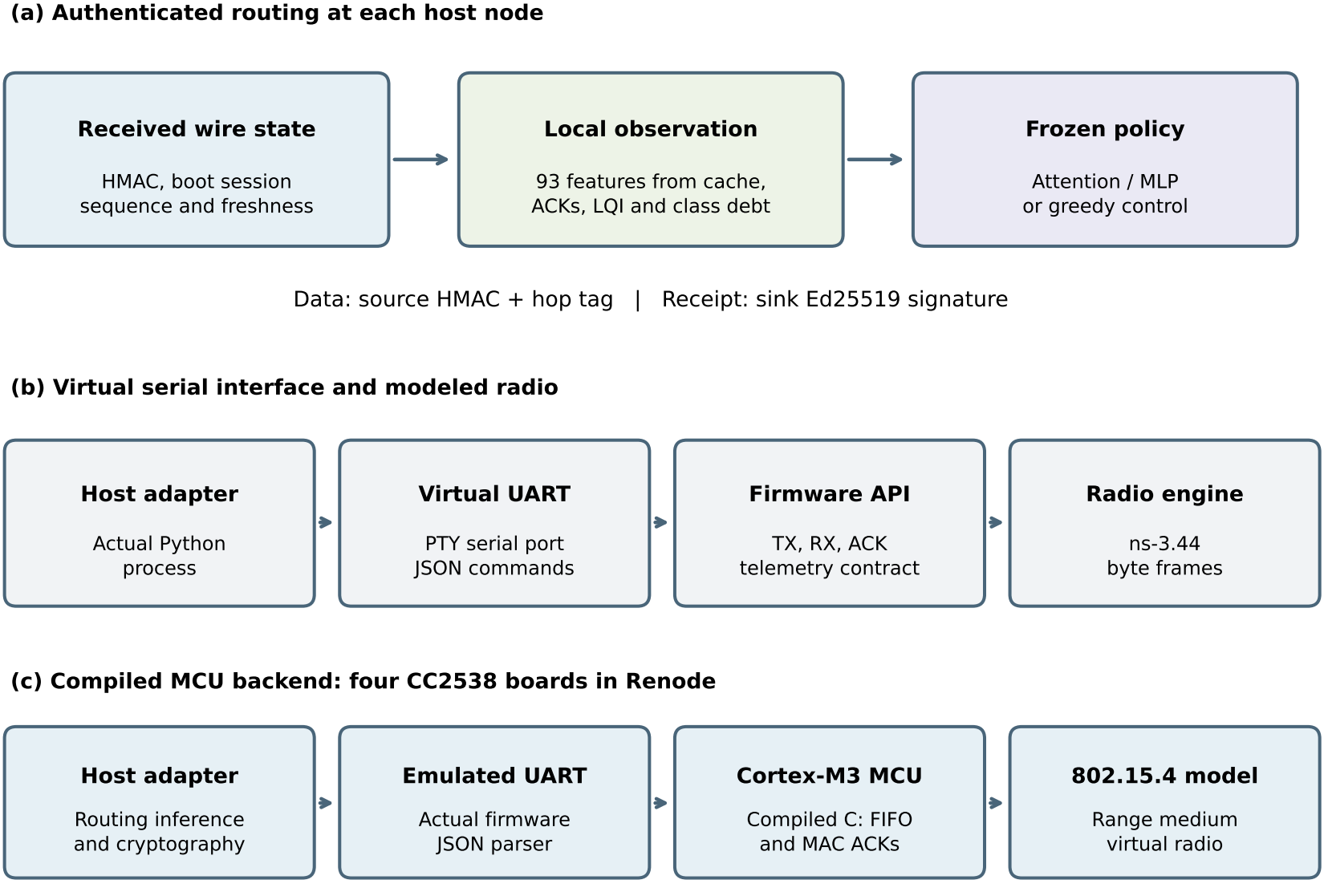
(a) Authenticated routing at each host node
Received wire state
HMAC, boot session
sequence and freshness
Local observation
93 features from cache,
ACKs, LQI and class debt
Frozen policy
Attention / MLP
or greedy control
Data: source HMAC + hop tag | Receipt: sink Ed25519 signature
(b) Virtual serial interface and modeled radio
Host adapter
Actual Python
process
Virtual UART
PTY serial port
JSON commands
Firmware API
TX, RX, ACK
telemetry contract
Radio engine
ns-3.44
byte frames
(c) Compiled MCU backend: four CC2538 boards in Renode
Host adapter
Routing inference
and cryptography
Emulated UART
Actual firmware
JSON parser
Cortex-M3 MCU
Compiled C: FIFO
and MAC ACKs
802.15.4 model
Range medium
virtual radio
Virtual backends; no physical RF or calibrated power measurement